

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1507

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct

I _Narmatha J (192521341)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _NARMATHA J

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

ANSWERS:

1. Create a Simple Cloud Software Application for Hospital Information System for SIMATS Hospital using Zoho Creator to Demonstrate SaaS

Aim:

To develop a simple Hospital Information System for SIMATS Hospital using Zoho Creator and demonstrate the Software as a Service (SaaS) model.

Software / Tools Required:

- Zoho Creator
- Web Browser
- Internet Connection
- Laptop/Desktop

Theory:

Software as a Service (SaaS) is a cloud computing model where applications are hosted on the cloud and accessed through the internet.

In this project, a Hospital Information System is developed using Zoho Creator. The application stores patient details, attender information, doctor details, disease information, and treatment status. The records are securely stored in the cloud and can be accessed and managed easily by hospital staff.

Algorithm:

1. Start.
2. Open the Hospital Information System.
3. Enter patient details.
4. Enter attender information.
5. Select the doctor and department.
6. Enter disease details.
7. Select the admission date and treatment status.
8. Submit the form.
9. Store the patient information in the cloud database.
10. View and manage patient records.
11. Stop.

Form Fields

Field Name	Field Type
Patient ID	Auto Number
Patient Name	Name
Age	Number
Gender	Drop-down
Address	Address
Phone	Phone

Email	Email
Attender Name	Name
Attender Phone	Phone
Doctor Name	Drop-down
Department	Drop-down
Disease	Multi Line
Date of Admission	Date
Blood Group	Drop-down
Treatment Status	Drop-down

Procedure:

- Log in to Zoho Creator.
- Create a new application named SIMATS Hospital Information System.
- Create a form named Patient Information.
- Add all the required fields.
- Configure the drop-down lists.
- Save the form.
- Enter sample patient details.
- Click Submit.
- Verify that the patient information is stored successfully in the cloud database.
- View the records using the generated report.

Output:

SIMATS Hospital Information Sys...
Patient Information

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Time

Drop Down

Radio

Multi Select

Checkbox

Patient ID

Patient Name

Age

Gender

Address

Phone

Email

Attender Name

Attender Phone

Doctor Name

Department

Disease

Date of Admission

Blood Group


SIMATS Hospital Information System

Development

Patient Information

Patient Information Report

Patient Information

Patient Name

Ravi

Kumar

First Name

Last Name

Age

35

Gender

Male

✕

▼

Address

Address Line 1

Address Line 2

Chennai

TamilNadu

City / District

State / Province

600099

India

✕

▼

Postal Code

Country

Phone

+91

9876543210

Email

ravi@gmail.com

Attender Name

Suresh

Kumar

First Name

Last Name

Attender Phone

+91

9876501234

Doctor Name

Dr. Priya

✕

▼

Department

General Medicine

✕

▼

Disease

Viral Fever

Date of Admission

01-Aug-2026

Blood Group

O+

✕

▼

Treatment Status

Under Treatment

✕

▼

Submit

The screenshot displays the SIMATS Hospital Information System interface. On the left is a dark sidebar with navigation options: 'Patient Information' (selected), 'Patient Information', and 'Patient Information Report'. The main area is titled 'Patient Information Report' and shows a table with one patient entry: Patient ID 1, Name Ravi Kumar, Age 35, Gender Male, and Phone +919876543210. To the right of the table is a detailed form for Patient ID 1, containing fields for Patient Name, Age, Gender, Phone, Email, Address, Attender Name, Attender Phone, Disease, Date of Admission, Blood Group, Department, Doctor Name, and Treatment Status.

Patient ID	Patient Name	Age	Gender	Phone
1	Ravi Kumar	35	Male	+919876543210

Patient ID	1
Patient Name	Ravi Kumar
Age	35
Gender	Male
Phone	+919876543210
Email	ravi@gmail.com
Address	Chennai, TamilNadu, 600099, India
Attender Name	Suresh Kumar
Attender Phone	+919876501234
Disease	Viral Fever
Date of Admission	01-Aug-2026
Blood Group	O+
Department	General Medicine
Doctor Name	Dr. Priya
Treatment Status	Under Treatment

Result:

The Hospital Information System was successfully developed using Zoho Creator. The application allows users to manage patient, attender, doctor, and treatment details securely in the cloud. This demonstrates the implementation of the Software as a Service (SaaS) model using Zoho Creator.

2. Create a Simple Cloud Software Application for Online Supermarket using Zoho Creator to Demonstrate SaaS

Aim:

To develop a simple Online Supermarket System using Zoho Creator and demonstrate the Software as a Service (SaaS) model.

Software / Tools Required:

- Zoho Creator
- Web Browser
- Internet Connection
- Laptop/Desktop

Theory:

Software as a Service (SaaS) is a cloud computing model where applications are hosted on the cloud and accessed through the internet.

In this project, an Online Supermarket System is developed using Zoho Creator. The application allows customers to enter their details, select items, specify quantity, and place orders. An item classification form is used to manage products according to their categories. All information is stored in the cloud.

Algorithm:

1. Start.
2. Open the Online Supermarket System.
3. Enter customer details.
4. Select the item and category.
5. Enter the quantity.
6. Check the item availability.
7. Calculate the total amount.
8. Enter payment details.
9. Submit the order.
10. Store the order information in the cloud.
11. Stop.

Form Fields:**Customer Order Form:**

Field Name	Field Type
Order ID	Auto Number
Customer Name	Name
Address	Address
Phone	Phone
Email	Email

Item Name	Drop-down
Item Category	Drop-down
Quantity	Number
Price	Currency
Total Amount	Currency
Order Date	Date
Payment Status	Drop-down

Item Classification Form:

Field Name	Field Type
Item ID	Auto Number
Item Name	Single Line
Category	Drop-down
Brand	Single Line
Price	Currency
Available Quantity	Number
Unit	Drop-down
Item Status	Drop-down

Procedure:

- Log in to Zoho Creator.
- Create a new application named Online Supermarket System.
- Create a Customer Order form.
- Add customer and order details.
- Create an Item Classification form.
- Add item name, category, price, quantity, unit, and status fields.
- Configure the required drop-down values.
- Save the forms.
- Enter sample customer and item details.
- Submit the order.
- Verify that the records are stored successfully in the cloud database.
- View the records using reports.

Output:

Online Supermarket Manageme...
Customer Order

Basic Fields



Name



Email



Address



Phone



Single Line

I

Multi Line

123

Number



15

Date



Time

Drop Down



Radio



Multi Select



Order ID



Customer Name



Address



Phone



Email



Item Name



Item Category

123

Quantity

\$

Price

\$

Total Amount



Order Date



Payment Status



Online Supermarket Manageme...
Item Classification

Basic Fields



Name



Email



Address



Phone



Single Line

I

Multi Line

123

Number



15

Date



Time

Drop Down



Radio



Multi Select



Item ID



Item Name



Category



Brand

\$

Price

123

Available Quantity



Unit



Item Status



Online Supermarket Management System

Development

OSM Online Supermarket Management...

Customer Order

Customer Order

Customer Order Report

Item Classification

Customer Order

Customer Name

Narmatha

J

First Name

Last Name

Address

No.163,BRG Nagar,

Address Line 1

3rd Streed,Kolathur

Address Line 2

Chennai

TamilNadu

City / District

State / Province

600099

India

Postal Code

Country

Phone

+91

9940200648

Email

narmatha@gmail.com

Item Name

Milk

Item Category

Dairy Products

Quantity

4

Price

22.0

Total Amount

88.0

Order Date

01-Aug-2026

Payment Status

Paid

Submit

Online Supermarket Management System

Development

OSM Online Supermarket Management...

Customer Order

Item Classification

Item Classification

Item Classification Report

Item Classification

Item Name

Milk

Category

Dairy Products

Brand

Aavin

Price

22.0

Available Quantity

4

Unit

Packet

Item Status

Available

Submit

Online Supermarket Management System | Development | View as: -MySelf- | Trial expires in 15 days | Upgrade | Edit this application | Help

Customer Order Report

Order ID	Customer Name	Address
1	Narmatha J	No.163,BRG Nagar,, 3rd Streed,Kolathur, Chennai, Tam

Order ID	1
Customer Name	Narmatha J
Address	No.163,BRG Nagar,, 3rd Streed,Kolathur, Chennai, TamilNadu, 600099, India
Phone	
Item Category	Dairy Products
Item Name	Milk
Email	192521341.simats@saveetha.com
Price	₹ 22.00
Total Amount	₹ 88.00
Quantity	4
Order Date	01-Aug-2026
Payment Status	Paid

[Add a comment](#)

Online Supermarket Management System | Development | View as: -MySelf- | Trial expires in 15 days | Upgrade | Edit this application | Help

Item Classification Report

Item ID	Item Name	Category	Brand
1	Milk	Dairy Products	Aavin

Item ID	1
Item Name	Milk
Category	Dairy Products
Brand	Aavin
Price	₹ 22.00
Available Quantity	4
Unit	Packet
Item Status	Available

[Add a comment](#)

Result:

The Online Supermarket System was successfully developed using Zoho Creator. The application allows customers to manage their details, select items, specify quantities, and place orders. The item classification form helps organize supermarket products. The data is stored in the cloud, demonstrating the SaaS model.

3. Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Aim:

To demonstrate virtualization by using VMware Workstation, a Type-2 hypervisor, and configure an Ubuntu Virtual Machine with 3 CPU cores, 10 GB RAM, and 10 GB storage.

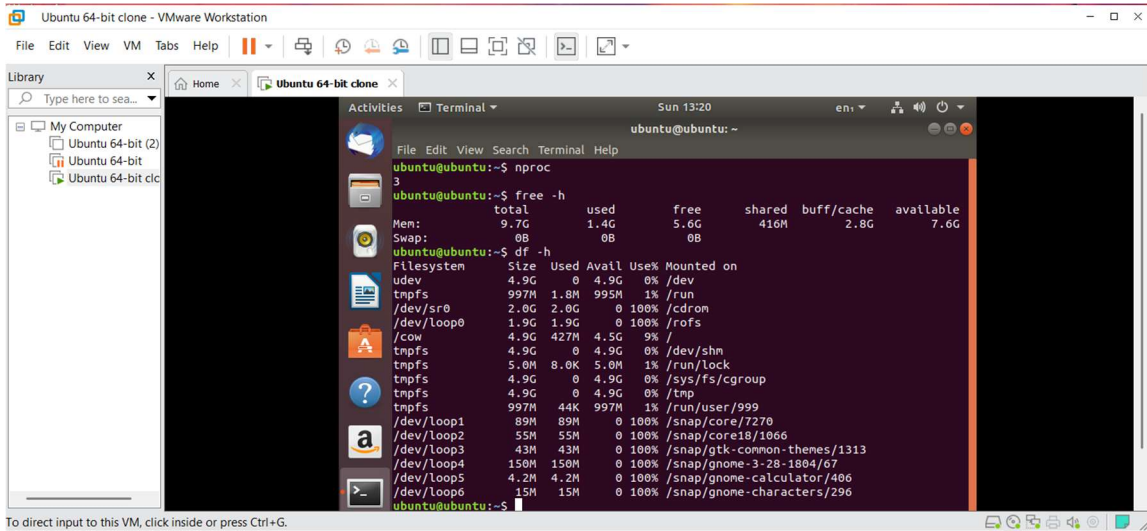
Software / Tools Required:

- VMware Workstation
- Ubuntu 64-bit Virtual Machine
- Windows Host Operating System
- Laptop/Desktop

Procedure:

- Open VMware Workstation and select the Ubuntu VM.
- Go to VM → Settings.
- Set RAM = 10 GB and CPU = 3 cores.
- Check the virtual disk storage.
- Start the Ubuntu VM and verify the configuration.
- Run the VM successfully.

Output:



Ubuntu 64-bit clone - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- Ubuntu 64-bit (2)
- Ubuntu 64-bit
- Ubuntu 64-bit clc

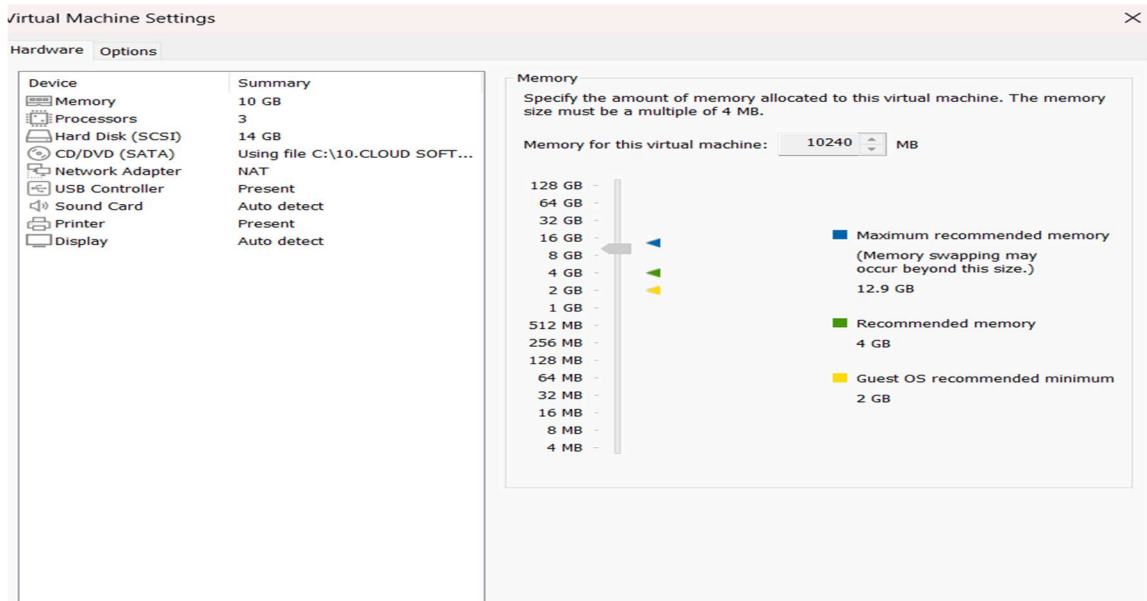
Activities Terminal

Sun 13:20 en

ubuntu@ubuntu: ~

```
ubuntu@ubuntu:~$ nproc
3
ubuntu@ubuntu:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           9.7G        1.4G        5.6G        416M        2.8G        7.6G
Swap:          0B           0B           0B
ubuntu@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            4.9G   0     4.9G   0% /dev
tmpfs           997M   1.8M 995M   1% /run
/dev/sr0        2.0G   2.0G   0 100% /cdrom
/dev/loop0      1.9G   1.9G   0 100% /rofs
/cow            4.9G  427M   4.5G   9% /
tmpfs           4.9G   0     4.9G   0% /dev/shm
tmpfs           5.0M   8.0K  5.0M   1% /run/lock
tmpfs           4.9G   0     4.9G   0% /sys/fs/cgroup
tmpfs           4.9G   0     4.9G   0% /tmp
tmpfs           997M   44K  997M   1% /run/user/999
/dev/loop1      89M    89M   0 100% /snap/core/7270
/dev/loop2      55M    55M   0 100% /snap/core18/1066
/dev/loop3      43M    43M   0 100% /snap/gtk-common-themes/1313
/dev/loop4      150M   150M   0 100% /snap/gnome-3-28-1804/07
/dev/loop5      4.2M   4.2M   0 100% /snap/gnome-calculator/406
/dev/loop6      15M    15M   0 100% /snap/gnome-characters/296
ubuntu@ubuntu:~$
```

To direct input to this VM, click inside or press Ctrl+G.



Virtual Machine Settings

Hardware Options

Device Summary

Device	Summary
Memory	10 GB
Processors	3
Hard Disk (SCSI)	14 GB
CD/DVD (SATA)	Using file C:\10.CLOUD SOFT...
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: 10240 MB

128 GB -
64 GB -
32 GB -
16 GB -
8 GB -
4 GB -
2 GB -
1 GB -
512 MB -
256 MB -
128 MB -
64 MB -
32 MB -
16 MB -
8 MB -
4 MB -

Maximum recommended memory (Memory swapping may occur beyond this size.) 12.9 GB

Recommended memory 4 GB

Guest OS recommended minimum 2 GB

Result:

The Ubuntu VM was successfully configured and executed using VMware Workstation Type-2 Hypervisor with 3 CPU cores, 10 GB RAM, and the required storage.

4. Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Aim:

To demonstrate virtualization using VMware Workstation Type-2 Hypervisor by creating a snapshot of an Ubuntu VM and restoring the VM to its previous version.

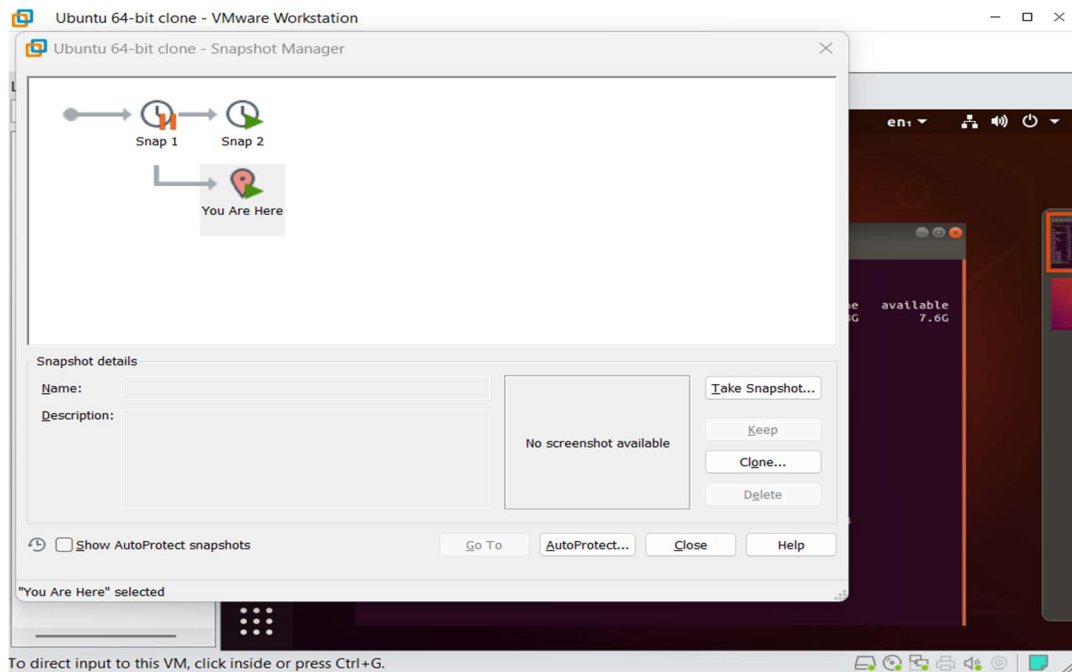
Software / Tools Required:

- VMware Workstation
- Ubuntu Virtual Machine
- Windows/Linux Host OS
- Laptop/Desktop

Procedure:

- Open VMware Workstation and start the Ubuntu VM.
- Make a small change in Ubuntu, such as creating a new folder or text file.
- Shut down the VM.
- In VMware, select VM → Snapshot → Take Snapshot.
- Give the snapshot a name such as Before Change.
- Start the Ubuntu VM and make another change.
- Shut down the VM.
- Select VM → Snapshot → Snapshot Manager.
- Select the Before Change snapshot.
- Click Go To / Restore.
- Start the Ubuntu VM and check whether it has returned to the earlier state.

Output:



Result:

A snapshot of the Ubuntu VM was successfully created using VMware Workstation. The VM was restored to its previous state successfully, demonstrating VM snapshot and recovery.